

Visible Spectrometer

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The objective of this lab is to evaluate the Rydberg constant and verify the equations that relate electron transitions in an atom to a discrete energy level in the Balmer ($n_f = 2$) spectral series. We do so by utilizing a reference helium discharge tube of known spectral emission wavelengths incident on a diffraction grating to calibrate angular offsets and uncertainties within our apparatus. After this, we observe the spectral emission of a hydrogen discharge tube, recording diffraction angles to determine the wavelength, initial quantum number, and best fit Rydberg constant to satisfy theoretical equations. After doing so, statistical analysis was done on the datasets to estimate the validity and error of the experiment itself.

1. INTRODUCTION AND THEORY

1.1. Rydberg Constant

The Rydberg constant has proven to be a reliable link between an atom's internal energy structure with respect to quantum number of electron transitions. It derives from coulomb electric potential of electrons and their kinetic energy from orbiting the nucleus (Eq. 1). In the Bohr model, these radii are defined from $n=1,2,3...$ with $n=1$ being the closest to the nucleus. It also takes into account the kinetic energy of the electron as it orbits the nucleus.

It relies on fundamental constants such as the reduced mass of an atom (Eq. 7), charge of an electron, permittivity of free space, Planck's constant.

$$E_n = K_n + U_n \quad (1)$$

$$E_n = \frac{\mu e^4}{8\epsilon_0 h^2 n^2} - \frac{\mu e^4}{4\epsilon_0 h^2 n^2} = -\frac{\mu e^4}{8\epsilon_0 h^2 n^2} \quad (2)$$

$$\Delta E_n = \frac{\mu e^4}{8\epsilon_0 h^2} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \quad (3)$$

By understanding the discrete energy levels that can be created from electron transitions, one can predict the energy of the photons emitted as a byproduct. Each element displays its own spectral series of discrete wavelengths, and this equation is powerful in the sense that it can use these series to describe the internal structure of the atom itself.

$$\Delta E_n = \frac{hc}{\lambda} \quad (4)$$

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \quad (5)$$

$$R_H = \frac{\mu e^4}{8\epsilon_0^2 h^3 c} \quad (6)$$

Reduced mass itself is highly specific to the atoms electron and proton mass which differs vastly between elements.

$$\mu = \frac{m_e m_p}{m_e + m_p} \quad (7)$$

Another source of variability is its connection to permittivity of the space in which it's transitioning. This stems from the electric potential/kinetic energy inherent in electrons as they are positioned in space and move around it. Most of the time we assume the permittivity of a vacuum, but our experiment takes place in air which notably differs in value.

This adjustment in energy due to the permittivity of air can be accounted for after the fact with the following equation:

$$\lambda_{air} = \frac{\lambda_{vacuum}}{n} \quad (8)$$

The wavelength of the photon emitted is slightly smaller than that of a vacuum, which makes sense because the speed of the photon itself decreases from the constant $c = 3 \times 10^8$ m/s of a vacuum.

It's important to note that the equation being used to determine wavelengths in this lab rely solely on Coulomb potential and kinetic energy. It fails to account for more subtle contributions to total energy levels, such as orbital angular momentum which can slightly raise or lower predicted values in what's known as the fine structure. Another subtle effect comes from nucleic forces, which causes additional variations in what's known as the hyperfine structure. Both of these are negligible in our current apparatus, but important to be aware of.

1.2. Trigonometric Equations of Diffraction

The angles measured in this experiment must be put into context of the photon rays emitted from the discharge tubes and their incident or refracted angles with

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respect to the diffraction grating. A visual of this path is included in figure 1.

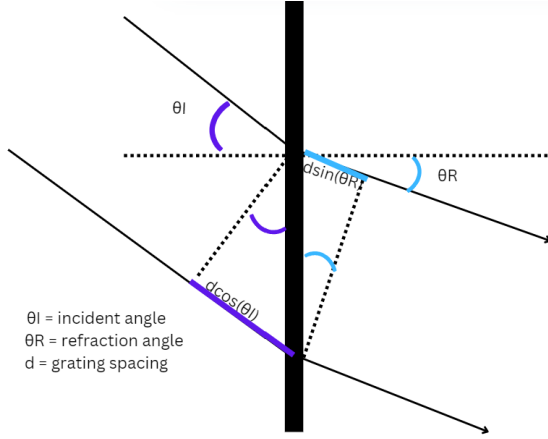


FIG. 1: Depiction of $d\sin\theta_i$ and $d\cos\theta_r$ used to find additional distance traveled by emitted rays incident on diffraction grating. The difference should be of an order $m\lambda$ in order to align in-phase with one another and constructively interfere into spectral lines.

$$m\lambda = d[\sin\theta_r - \sin\theta_i] \quad (9)$$

Eq. 9 represents the additional distance in optical path length from one ray to the other. The condition for a spectral line to appear is that constructive interference of these two rays must occur. This happens when both arrive in phase with one another, and any additional distance of one is on the order of a full wavelength times any integer m .

This equation can easily be converted to more readily measured variables of H_r (refracted angle) and D_c / D_s / D_0 (systematic / setup constants). A more detailed approach is proven in Appendix 1.1.

$$m\lambda = D_s \sin H_r + D_c \cos H_r + D_0 \quad (10)$$

2. EXPERIMENT

2.1. Equipment

The equipment being used consists of a collimator, telescope, diffraction grating, hydrogen/helium discharge tube, sodium lamp, and adjustable camera as show in Fig. 2. The collimator is used to reduce the amount of light coming into the diffraction grating in either the horizontal or vertical direction, roughly keeping the incident rays parallel to one another. It can be widened or thinned depending on purpose of experimental section. The telescope is used to visualize spectral lines and can be rotated along the table itself, which corresponds with different vernier scale readings of diffracted angles.

The diffraction grating is placed upon a table with leveling screws to fine tune the alignment of incident and refracted rays with the plane of the collimator and telescope highlighted in the following steps.

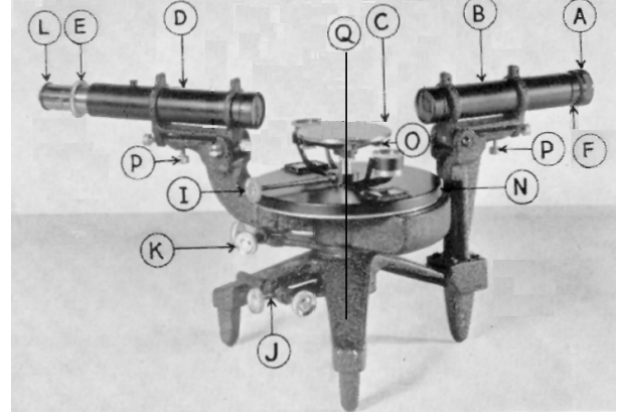


FIG. 2: The spectrometer used in this lab, with collimator to the right, telescope to the left, and grating plate at the center. [1]

2.2. Alignment of Apparatus

1. Take spectrometer to hallway and focus telescope lens on bulletin board at the end of the hallway to remove parallax - this is a good reference to infinite distance
2. Place light bulb in front of collimator, rotating the telescope to see directly into the slit. Move collimator tube until a sharp horizontal slit comes into focus on the telescope. Adjust the entrance slit for a clear, yet reasonably bright slit.
3. Ensure collimator and telescope optical axes are aligned by adjusting the telescope leveling screws until cross hairs are centered on the slit.
4. Mount grating assembly to the mounting post centered between the collimator and telescope. Check that the grating face itself bisects the plate so that it lies on the diameter with the tilt adjustment screw. Make sure that it's installed at a height level with collimator and telescope.
5. Rotate the telescope about 90 degrees to the collimator until it's reflection is visible Adjust leveling screws on grating plate to center reflected image on cross hairs. Because we know the collimator and telescope are at the same height from step 4, this ensures the grating normal is perpendicular to the main rotation axis and parallel to the two's line of height. If it was not, the reflection would bounce off at an angle equal to its incidence in the z-axis, either above or below the original image it was centered at in step 4.
6. Rotate the telescope an additional 45 degrees or until it's roughly normal with the grating surface. Turn on a flashlight and place it at the telescopes' side hole which

will produce a circular reflection of light. Adjust the table leveling screw until cross hairs of the original and reflected image are aligned. This establishes a parallel telescope axis to the grating normal (autocollimation).

7. Remove table with grating so that the telescope can look directly at the collimator. Adjust the collimators leveling screw until the cross hairs are once again aligned with the slit.

8. Repeat steps 4-6 as needed to improve alignment.

9. Place the table with grating back in center of apparatus.

10. Adjust vernier scale by loosening and re-tightening table base locking screw so that vernier scales A and B are roughly 90 degrees to the collimator. This ensures an easy reading of angles during the experiment.

11. Rotate table base of grating approximately 7 degrees and make sure the hollow side faces collimator. Check that the center of the collimator is roughly centered with the grating face so that maximum light is transferred through.

12. Go through autocollimation process again of shining light through the telescopes side hole, adjusting leveling screws as needed to align cross hairs to reflected image.

13. When rotating the telescope, a gradient of colors should be displayed. Adjust the inline leveling screw until this line is centered at the cross hairs for these large telescopic angles.

2.3. Calibration with Helium

14. Find the direction of the grating normal by recording angles A_n and B_n on the vernier scales at the point where autocollimation occurs.

15. Turn slit vertical and replace the light bulb with a helium discharge tube. Align the telescope cross hairs with the zero order image, recording respective A_i and B_i angular values.

17. Record all other A_n and B_n values for further orders, recording their wavelengths as [violet: 447.15 nm] [blue-green: 501.57 nm] [orange: 587.57 nm] [red: 667.82 nm]. Record estimated uncertainty in arc minutes as σ_r .

2.4. Hydrogen Measurements

18. Switch out helium tube for hydrogen, keeping it in close proximity with the entrance slit and confirming brightest possible image at the telescope end.

19. Record all A_n and B_n values of hydrogen discharge spectrum lines for orders 1-3. Use video camera to check for weaker spectral lines and apply larger σ_r uncertainties if necessary.

2.5. Resolving Power

20. Replace hydrogen discharge tube with sodium lamp (may take about 10 minutes to heat up). Find the sodium doublet lines in first order with a 0.6 nm difference in wavelength. Add auxiliary slit over the end of collimator closest to diffraction grating. Decrease it's width until the two lines are just past the point of being differentiated by the human eye. Measure this slit width and calculate the respective grating grooves illuminated. Check second order lines to see how they behave at larger angles.

*all steps in reference to lab manual [1]

3. RESULTS

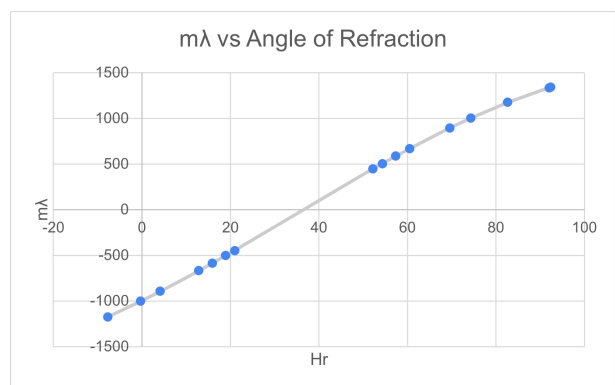


FIG. 3: Measured angle of refraction (H_r) in helium with regard to order and wavelength of ray. Wavelength was predetermined for helium spectrum and order was decided to be positive at $\sin\theta_r > \sin\theta_i$.

Figure 3 shows the relationship between the found angle of refraction at various spectral lines produced from the helium sample. While subtle, it produces a sinusoidal waveform - with a slight concave up to the left and concave down to the right. This agrees with theory A7 because $m\lambda$ is related to H_r by a combination of sine and cosine terms of the same behavior.

When fitting this data to Eq. A7, the following parameters were found to satisfy the relationship [$d = 1657.32$, $H_n = 0.5952$, $H_i = 0.6362$]. These agree with the original H_n and H_i radian values recorded at autocollimation and zero order images in steps 14-15 of Calibration [$H_{n\text{measured}} = 0.5998$, $H_{i\text{measured}} = 0.6369$].

From both the calibration fit and the Rydberg fit, we calculated higher χ^2 values than anticipated. In the calibration, using an angular uncertainty of 2 arcmins we obtained a high χ^2 value of 120. This could be due to underestimating the variation in measured angular values for H_r . It is more likely that we had an experimental angular uncertainty of approximately

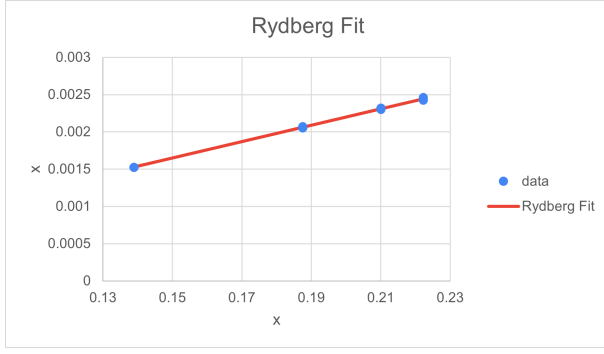


FIG. 4: Relationship between x and y variables where $y = \frac{1}{\lambda}$ and $x = \frac{1}{n_f^2} - \frac{1}{n_i^2}$. The slope of the relation is theoretically predicted to be Rydberg constant, referencing Eq. 5.

5 arc minutes, where our χ^2 value was minimized. Our residuals appeared to be reasonable - at < 5 nanometers from from predicted $m\lambda$. Additionally, neither the hydrogen or helium residuals showed a pattern in their magnitudes, indicating that no significant systematic errors contributed [5, 6]. The calibration values were in close range to theoretical values, with an evident sinusoidal relationship between $m\lambda$ and H_r . The Rydberg displayed a similar agreement with equations, outlining a linear relationship between x ($\frac{1}{n_f^2} - \frac{1}{n_i^2}$) and y ($\frac{1}{\lambda}$). The slope of this line is the experimental Rydberg value, at 0.01100 nm^{-1} . The widely accepted value for Rydberg constant for Hydrogen is 0.010968 nm^{-1} and at infinite mass, 0.010973 nm^{-1} . While our experimental Rydberg is above this value by approximately 0.029 percent, it is enough to prevent our ability to differentiate a hydrogen constant from the infinite mass constant.

The medium in which we took experimental measurements has an index of refraction higher than that of a vacuum, producing a reduced wavelength evident in Eq. 8. This would subsequently lead to a perceived higher energy level than previously calculated on the spectral lines. While it does have some effects, the magnitude of variation is roughly < 1 nanometer (based around sample calculation of 650 nm and $n_{air} = 1.0002$) [2]. Comparing this to our $\Delta m\lambda$ values of the calibration measurements compared to predicted- which are on the magnitude of $1\text{-}4 \text{ nm}$ - the slight decrease in wavelength wouldn't be an issue.

4. CONCLUSIONS

In conclusion, this lab effectively evaluated the Rydberg constant and verified the theoretical relationships governing electron transitions in hydrogen and helium. By calibrating our setup with known spectral lines from the helium discharge tube, we accurately measured the diffraction angles and determined the wavelengths emitted from the hydrogen discharge tube. Our experimental Rydberg constant of 0.01100 nm^{-1} closely aligns with accepted values, differing by only about 0.029 percent. This minor variation can likely be attributed to uncertainties in angular measurements and the refractive effects of air on the emitted wavelengths.

This experiment demonstrates the power of the Rydberg formula in linking atomic structure to spectral emissions. While we encountered some limitations due to experimental errors, the results are not far from the theoretical predictions of the Bohr model. Future experiments could go deeper into the impacts of fine and hyperfine structures.

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- [1] R. DeSerio, *Visible spectrometer manual*, http://http://www.phys.ufl.edu/courses/phy4803L/group_III/optical_spectroscopy/Spec.pdf (2022, year the manual was last updated), Phy 4803L, Department of Physics, University of Florida.
- [2] J. A. Stone and J. H. Zimmerman, *Engineering metrology toolbox* (2019), accessed: 2024-10-19, URL <https://emtoolbox.nist.gov/Wavelength/Documentation.asp>.

Appendix A: Auxiliary material

1. Derivation of linear regression of fitting parameters

$$m\lambda = d[\sin(H_r - H_n) - \sin(H_i - H_n)] \quad (\text{A1})$$

$$\sin(a - b) = \sin(a) \cos(b) - \cos(a) \sin(b) \quad (\text{A2})$$

$$m\lambda = d[\sin(H_r) \cos(H_n) - \cos(H_r) \sin(H_n) - \sin(H_i - H_n)] \quad (\text{A3})$$

Substituting the fitting parameters:

$$D_s = d \cos H_n \quad (\text{A4})$$

$$D_c = -d \sin H_n \quad (\text{A5})$$

$$D_0 = -d \sin(H_i - H_n) \quad (\text{A6})$$

Therefore:

$$m\lambda = D_s \sin(H_r) + D_c \cos(H_r) + D_0 \quad (\text{A7})$$

We can use this equation to model $m\lambda$ values as a form of linear regression with direct respect to measured angles from our apparatus (H_r).

2. Checking individual parameters

$$d = \sqrt{D_s^2 + D_c^2} \quad (\text{A8})$$

Substituting values for D_s and D_c

$$d = \sqrt{d^2(\cos^2(H_n) + \sin^2(H_n))}$$

knowing the identity:

$$\sin^2(\theta) + \cos^2(\theta) = 1,$$

we can simplify the d equation to be

$$d = \sqrt{d^2},$$

Confirming the previous equations for D_s and D_c relative to d.

$$H_n = -\tan^{-1}\left(\frac{D_c}{D_s}\right) \quad (\text{A9})$$

$$H_n = -\tan^{-1}\left(\frac{-d \sin(H_n)}{d \cos(H_n)}\right)$$

Therefore $H_n = \tan^{-1}(\tan(H_n))$

Lastly:

$$H_i = -\sin^{-1}\left(\frac{D_0}{\sqrt{D_s^2 + D_c^2}}\right) - \tan^{-1}\left(\frac{D_c}{D_s}\right) \quad (\text{A10})$$

$$H_i = -\sin^{-1}\left(\frac{D_0}{d}\right) + H_n$$

$$H_i = -\sin^{-1}\left(\frac{-d \sin(H_i - H_n)}{d}\right) + H_n$$

$$H_i = H_i - H_n + H_n$$

So all relationships are satisfied with this system of equations.

3. Calculating Jacobian

The b of d Jacobian matrix involved in calculating σ_d is obtained by taking the derivative of fitting parameters

from Eq. A4-A7 with respect to d, H_n , and H_i .

The results are as follows:

$$\frac{\partial D_0}{\partial d} = -\sin(H_i - H_n)$$

$$\frac{\partial D_s}{\partial d} = \cos(H_n)$$

$$\frac{\partial D_c}{\partial d} = -\sin(H_n)$$

$$\frac{\partial D_0}{\partial H_n} = d \cos(H_i - H_n)$$

$$\frac{\partial D_s}{\partial H_n} = -d \sin(H_n)$$

$$\frac{\partial D_c}{\partial H_n} = -d \cos(H_n)$$

$$\frac{\partial D_0}{\partial H_i} = -d \cos(H_i - H_n)$$

$$\frac{\partial D_s}{\partial H_i} = 0$$

$$\frac{\partial D_c}{\partial H_i} = 0$$

4. Residual Graphs

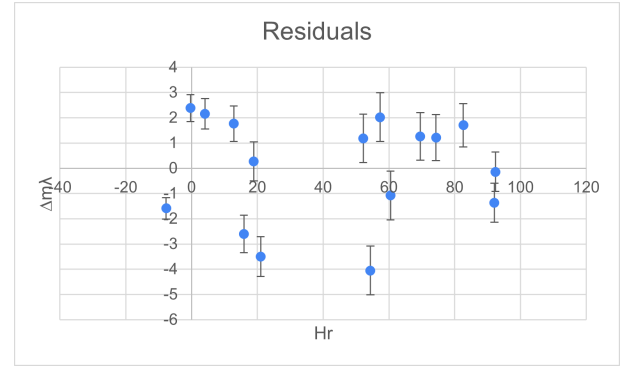


FIG. 5: The residuals gathered when comparing ideal $m\lambda$ values for helium to those calculated from fitting parameters.

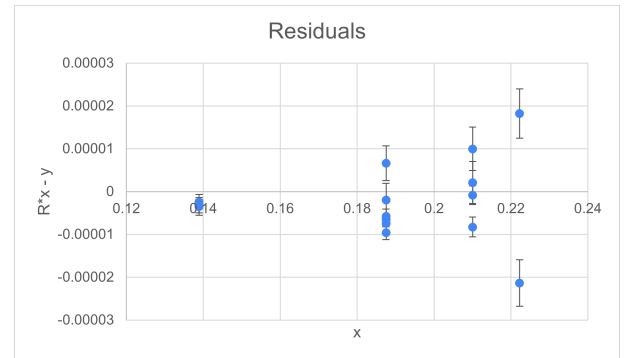


FIG. 6: The residuals gathered when comparing ideal $m\lambda$ values for hydrogen to those calculated from fitting parameters.